

TrainXcel Backend - Development Work Report

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1. Technologies & Architecture Used

- NestJS Framework (Core API & architecture)
- TypeScript (Strict compile-time type safety)
- TypeORM (Object-Relational Mapping for Postgres)
- Neon Serverless PostgreSQL (Cloud database environment)
- JWT Authentication (Deliverable via HTTP-only Cookies)
- class-validator & class-transformer (DTO-based input validation)

2. Summary of Work Accomplished

- **Custom Sequential ID System:** Built prefix-based auto-incrementing custom strings (TX-XXXX, CRS-XXXX, LES-XXXX) for public APIs, decoupling raw database primary key sequences.
- **Session-based JWT Auth:** Integrated HTTP-Only cookies to protect token storage from script hijacking, automatically binding user context (userId, role) to API requests.
- **Dynamic Progress Tracking:** Formulated double-precision completion rates for course enrollments, automatically calculating progress on lesson completions and additions.
- **Advanced Analytics:** Built 9 distinct manager dashboard endpoints to compile rolling monthly statistics, course leaderboards, at-risk user alerts, and recent timelines.
- **Unified Search Module:** Implemented dynamic fuzzy string search to find courses and employee profiles from a single unified route query.

3. Core API Endpoints & Aims

[POST] /auth/register — Creates profile & signs JWT session cookie
[POST] /auth/login — Validates credentials & sets session cookie
[POST] /auth/logout — Clears the JWT cookie session
[GET] /auth/profile — Gets profile of the currently logged-in user
[PATCH] /auth/users/:userId — Updates profile name, phone, and address
[POST] /courses — Creates new course (Admin/Employee only)
[PATCH] /courses/:courseId — Updates course details (Admin/Employee only)
[POST] /courses/:courseId/lessons — Adds a lesson (Admin/Employee only)
[POST] /courses/:courseId/enroll — Enrolls currently logged-in user in course
[POST] /courses/:courseId/lessons/:lessonId/complete — Completes lesson & updates progress
[GET] /courses/search/unified — Unified search for courses & employees
[GET] /courses/stats/dashboard — Computes overall statistics counters
[GET] /courses/stats/monthly-progress — Dynamic 12-month average progress curve
[GET] /courses/stats/at-risk — Flags users with progress below 15%
[GET] /courses/stats/recent-activity — Provides live chronological action feed